

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All source data was collected from seven US states and two Australian states during 2020–2021 [Q123], and geographic areas were selected for their fit to the four-domain taxonomy [Q124]. Tweet filtering depends on US/Australian city, town, and 'famous neighborhoods' lists [Q30] and on English-curated keyword lists [Q126]. The 1,000 clusters were hand-selected by the authors [Q52]. Ecuadorian crisis tweets reference parroquias, cantones, SNGR (formerly SGR) [WEB-11], IGEPN, INAMHI, Cruz Roja Ecuatoriana, volcano names (Cotopaxi, Tungurahua, Sangay), Kichwa toponyms, and operationally critical terms like 'damnificados' and 'albergues' — none of which are represented in the dataset. Elicitation A2 confirms analysts require precise anchoring to Ecuador-specific entities and that US-default references would be operationally meaningless. The Ecuador-specific UrbangEnCy dataset [WEB-17] exists but covers only binary classification, not timelines.

ID	Evidence
Q30	"The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest." (p.3)
Q52	"We, authors of this work, manually selected 1,000 clusters that contain enough tweets describing how a crisis event evolves, while specifying the "seed tweet" (i.e. the first observed post that describes the ongoing event) of each timeline." (p.5)
Q123	"We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria." (p.12)
Q124	"These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York." (p.13)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)
WEB-11	SNGR was formally renamed from SGR in 2024; both acronyms appear in tweets. (https://www.gestionderiesgos.gob.ec/wp-content/uploads/downloads/2024/05/Resol.SNGR-102-2024.pdf)
WEB-17	UrbangEnCy demonstrates feasibility of Spanish Ecuadorian Twitter capture but is binary-classification only. (https://www.sciencedirect.com/science/article/pii/S2352340920315729)

Strengths

- The benchmark provides a methodological template (semi-automated cluster-then-refine, seed-tweet specification) [Q27, Q52] that could in principle be re-applied to Ecuadorian Spanish content; the authors transparently document content-collection limitations [Q117] and explicitly acknowledge that replication for other languages/locations is hard [Q119].

ID	Evidence
Q27	"This section presents our semi-automated approach to collect CrisisLTLSum. We first extract clusters of tweets as noisy timelines and then refine them via human annotation to get clean timelines that only include non-redundant, informative, and relevant tweets." (p.3)

Q52	"We, authors of this work, manually selected 1,000 clusters that contain enough tweets describing how a crisis event evolves, while specifying the "seed tweet" (i.e. the first observed post that describes the ongoing event) of each timeline." (p.5)
Q117	"This study has some limitations on the dataset generation workflow. First, our noisy timeline collection process is not a comprehensive and exhaustive extraction to find all available information about a local crisis event. Here, our focus is to provide a representative dataset rather performing a comprehensive set containing all the local crisis events and their updates." (p.10)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Ecuador queries require Spanish-language dialectal knowledge, Ecuadorian administrative geography (24 provinces, ~221+ cantones, 1,500+ parroquias) [WEB-6, WEB-8], Kichwa/Shuar toponym recognition [WEB-1], institutional acronyms (SNGR/SGR, IGEPN, INAMHI, MSP), and crisis-specific Spanish vocabulary ('damnificado', 'albergue', 'lahares', 'desbordamiento'). None of this is present in the benchmark.

IC-2: Cultural sensitivity is misaligned: the dataset reflects US/Australian crisis-response cultural norms (FEMA-style agencies, US journalistic conventions). Ecuadorian humanitarian context emphasizes inter-agency coordination, indigenous community signals, and the 'damnificado' legal-operational concept absent from the benchmark.

IC-3: Western-specific knowledge is pervasive: location candidate lists [Q30], OSM-based place anchoring [Q36], and keyword curation from US/Australian news [Q126] all encode US/Australian assumptions that will not transfer.

IC-4: INSUFFICIENT DOCUMENTATION — the benchmark itself documents no regional annotators; for Ecuador deployment, expert annotators from Cruz Roja Ecuatoriana, UN OCHA Ecuador, and indigenous federations (CONAIE, ECUARUNARI) would be required.

IC-5: Content validity is fundamentally violated: the entire instance pool reflects US/Australian crisis content [Q123, Q69], with no Spanish, Latin American, or humanitarian-organizational content represented.

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Q30	"The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest." (p.3)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q69	"Out of these 1,000 annotated timelines (10,610 tweets) in CrisisLTLSum, 423 (42%) are about" (p.5)
Q123	"We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria." (p.12)
Q124	"These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York." (p.13)
Q126	"We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past." (p.13)

WEB-1	Kichwa orthography is contested with continued spelling variation; written use of indigenous languages outside schools is minimal. (https://books.openbookpublishers.com/10.11647/obp.0032/chapter11.html)
WEB-6	Ecuador has 24 provinces with distinct administrative hierarchy. (https://en.wikipedia.org/wiki/Provinces_of_Ecuador)
WEB-8	Ecuador has approximately 221 cantones and over 1,500 parroquias — administrative units absent from benchmark geography. (https://storyteller.travel/map-provinces-ecuador/)
WEB-11	SNGR was formally renamed from SGR in 2024; both acronyms appear in tweets. (https://www.gestionderiesgos.gob.ec/wp-content/uploads/downloads/2024/05/Resol.SNGR-102-2024.pdf)

Information Gaps

- OpenStreetMap coverage quality for rural Ecuadorian parroquias and Kichwa community names is unverified.

Requires Expert Verification

- Specific institutional Twitter/X handles for SNGR, IGEPN, INAMHI, and Cruz Roja Ecuatoriana for entity-linking.

Remediation

Gap 1: All instances are US/Australian English Twitter; no Ecuadorian geographic, institutional, or linguistic content.

Recommendation: Build an Ecuador-specific timeline corpus using UrbangEnCy [WEB-17] as a Spanish-tweet seed pool, extended with parroquia-level location candidates, SNGR/IGEPN/INAMHI keyword lists, and Kichwa toponym dictionaries curated with CONAIE/ECUARUNARI input.

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WEB-17	UrbangEnCy demonstrates feasibility of Spanish Ecuadorian Twitter capture but is binary-classification only. (https://www.sciencedirect.com/science/article/pii/S2352340920315729)